Basic End-to-End Pipeline Design

Background:

As a Data Engineer, you frequently encounter requests to retrieve data from external sources like API endpoints or cloud storage and ingest it into your storage for further analysis. This task is designed to evaluate your capability in implementing a basic end-to-end data pipeline. There is no single correct solution; candidates are encouraged to use any programming language and framework of their choice.

Task Description:

Choose either CASE A or CASE B; **implementation of both cases is not required**.

Please answer the following questions based on the data from the provided API. Design and implement an end-to-end pipeline to accomplish this. The implementation should encompass an ETL (Extract, Transform, Load) workflow for fetching raw data from API endpoints, processing and extracting relevant information, and loading the data into tables in a local database. Candidates are free to utilize any programming language and framework.

For sharing the result, create a GitHub repository for the tasks and share it with us when you're ready. Ensure that the implementation can be executed error-free, at least in your development environment. If it's possible, deployment of the implementation as a service would be beneficial.

Result:

In addition to answering the questions, you must generate:

At least **three different types of visualizations** based on aggregated data.

Requirements:

- Visualizations can be produced using code (e.g., Python: matplotlib, seaborn, Plotly, dashboard frameworks) or via enterprise BI tools such as PowerBI or Looker.
- Charts must be based on processed data (not raw API dumps)
- Results must be reproducible
- Clear labels, titles, and ordering are expected

CASE A: Open Brewery Data API

API Documentation: Open Brewery Data API Documentation

1. Which state in the USA has the most microbreweries?
2. What are the top 5 states in the USA with the most microbreweries?
3. How many brewpub breweries are there in the state or province of Incheon, South Korea?
4. (Bonus + Optional) How would you analyze the phone number patterns with cities in Korea based on the information given by the API?

CASE B: Zelda API

API Documentation: Zelda API Documentation

1. Which staff member worked on the most games?
2. List the monsters that appear in at least 3 games.
3. List items with their descriptions ordered by the number of games in which they appeared, starting with the highest number of games.

Feel free to contact us whenever you confront any problems and questions. :-)